# Ghulam Ishaq Khan Institute (GIKI)	
Assignment 3 + Presentation	
Subject: Computer Architechture	**Course Code:** CS-361 Fall 25
Class: CS 33	
Instructor: Waheed Ahmad	**Total Marks:** 10 (4-6% weightage)

Assignment Instructions: Device Architecture Analysis Presentation

Course: CS361 – Computer Architecture
Instructor: Waheed Ahmad
Task: Analyze the architecture and performance of your personal device (phone or laptop)
Deliverable: A 5–7 slide, 3–5 minute presentation
Weightage: 4-6%

Group Work Requirement

- **Form a group of 2–3 members.**
- **Fill the following form to get your presentation slot:**

  Presentation slots CS361.xlsx

- **Only one submission per group on Teams.**
- **Each group must upload there presentations on Teams as well.**

Compare with a State-of-the-Art Architecture in Your Category

You *must* compare your chosen device to one current state-of-the-art CPU/GPU/SoC in the same category. Examples:

- If you choose an Android phone → compare with Snapdragon 8, mediaTek, or Tensor Gx
- If you choose an Intel laptop → compare with Intel 14th-gen, AMD Ryzen 7000, or Apple Mx

- If you choose an Apple device → compare with latest apple chipset Intel/AMD laptop chips or Snapdragon X Elite

This comparison must appear in:

- Slide 5 (Benchmark comparison)
- And optionally in Slides 2–4 (ISA, ILP, TLP comparisons)

Objective

This assignment helps you connect course concepts ISA, pipelining, parallelism, memory hierarchy with real hardware.

By the end, you will be able to:

1. Identify and contrast major ISA philosophies (ARM, x86, RISC-V).
2. Find ILP and pipelining techniques (superscalar, OoOE) in modern CPUs.
3. Evaluate TLP and memory hierarchy of a real device.
4. Measure and analyze performance (CLO-1, CLO-2, CLO-3).

Task Instructions

1. Choose one device: smartphone, personal laptop, or gaming console.
2. Research your architecture using deep technical sources
   (Tom's Hardware, AnandTech, WikiChip, official white papers).
3. Your presentation must cover the following:

A Note on Class Participation

Asking questions during peer presentations is encouraged.
In borderline grade cases (e.g., 89.5), active participation may be used to round up a student's grade. This is entirely at the instructor's discretion.

Slide Outline (5–7 Slides)

Slide 1 – Title Slide

- Name
- Roll number
- Devices name & processor

Slide 2 – CPU Architecture & ISA

- CPU name & ISA (x86-64, ARMv8, etc.)
- ISA philosophy: CISC or RISC?
- Contrast with RISC-V's open ISA philosophy
- (Maps to: Weeks 1–2)

Slide 3 – ILP, Pipelining & DLP [UPDATED]

- Evidence of ILP:
    - Superscalar width
    - Out-of-Order Execution
    - Branch prediction features
- DLP: SIMD extensions (AVX, NEON, etc.)
- (Maps to: Weeks 4–7)

Slide 4 – TLP & Memory Hierarchy

- Core count, performance vs. efficiency cores
- Threads (e.g., SMT/Hyper-Threading)
- Cache sizes (L1/L2/L3)
- RAM type & speed
- Storage type (NVMe, UFS, etc.)

- (Maps to: Weeks 8–15)

Slide 5 – Performance & Benchmarks [UPDATED WITH STATE-OF-THE-ART COMPARISON]

- Include benchmark scores (Geekbench, PassMark, Cinebench, etc.):
  - Single-core
  - Multi-core
- Compare with one state-of-the-art competing device in your category (current highest performing snapdragon or media tek, apple, Intel vs. AMD, etc.).
- Explain *why* one device is faster (link to Slides 3 & 4).
- **(Maps to: Week 3 – Performance)**

Slide 6 – Power & Energy Efficiency

- Battery life or CPU TDP
- Efficiency core architecture
- Energy-saving mechanisms
- (Maps to: Week 2)

Slide 7 – Summary

- **2**–3 main takeaways
- Major strength
- Biggest bottleneck
- Any surprising findings (what surprised you)

Presentation & Evaluation

Duration: 3–5 minutes
Submission: PowerPoint / PDF on Teams (one per group)

Evaluation Rubric (10 Marks)

Criteria	Marks
Technical Depth	**4**
Performance Analysis	**3**
Clarity & Delivery	**2**
Summary & Insights	**1**
Total	**10**

NEW NOTE (Important)

You will/may/can also be asked questions on the *concepts themselves* during the presentation.
This means that in addition to explaining your device, you must understand the underlying course topics such as:

- ISA types (ARM vs x86 vs RISC-V)
- Pipelining, hazards, OoOE
- Superscalar execution and ILP
- SIMD / DLP
- TLP (cores, threads, SMT)
- Cache hierarchy
- Memory bandwidth vs latency
- Benchmark interpretation

Your conceptual understanding will be evaluated through follow-up questions.